Performance Evaluation of HTTP/1.1, HTTP/2 and HTTP/3 with AI-Assisted Analysis

Abstract

Modern web applications rely on HTTP protocols to transfer data between clients and servers. HTTP/1.1, HTTP/2 and HTTP/3 use different communication mechanisms, which may result in different network performance. This study presents an experimental comparison of HTTP/1.1, HTTP/2 and HTTP/3 using the same web server and a single HTML document request.

Each protocol was tested repeatedly under the same experimental conditions. A total of 1000 measurements were collected for each protocol. Several performance metrics were recorded, including response time, Time To First Byte (TTFB), Round Trip Time (RTT), jitter, packet loss, data size, throughput and request success rate. The collected data was organised and analysed using Excel.

The purpose of this study is to compare the practical performance of the three HTTP protocols and investigate how their performance differs under the same testing conditions. The results show differences between the protocols across several performance metrics. A small AI-assisted analysis was also conducted using ChatGPT and DeepSeek to examine whether AI models could identify the protocol with the lowest measured response time. The results showed different prediction accuracy across the two testing rounds. These results suggest that AI can provide supplementary analysis, but its predictions should still be validated against direct experimental measurements.

Keywords: HTTP/1.1, HTTP/2, HTTP/3, QUIC, Web Performance, Response Time, TTFB, AI-Assisted Analysis.

1. Introduction

The Hypertext Transfer Protocol (HTTP) is one of the main protocols used for communication between web clients and servers. When users access a website, HTTP is responsible for transferring requests and responses between the client and the server. As modern websites become more complex, network performance and response time have become increasingly important for web applications.

HTTP has developed through several major versions. HTTP/1.1 has been widely used for many years and provides persistent connections. However, it does not provide native multiplexing for multiple requests. HTTP/2 introduced multiplexing, allowing multiple streams to be transferred through a single TCP connection. HTTP/3 uses the QUIC transport protocol instead of TCP and provides a different approach to connection and stream management.

Although newer HTTP versions provide improvements at the protocol level, their actual performance may not always be the same in different environments. Network conditions, server response time, connection behaviour and the type of workload can affect the measured results. Therefore, experimental testing is useful for understanding the practical performance of different HTTP protocols.

In this study, HTTP/1.1, HTTP/2 and HTTP/3 are tested using the same website and testing environment. This study focuses on the performance of a single HTML document request. Each protocol is tested 1000 times to collect a larger set of experimental data.

Several performance metrics are collected during the experiments, including response time, TTFB, RTT, jitter, packet loss, data size, throughput and success rate. The aim of this study is to compare the performance of HTTP/1.1, HTTP/2 and HTTP/3 under the same experimental conditions and to provide a more detailed view of their performance differences.

A small AI-assisted analysis was also conducted using ChatGPT and DeepSeek to examine whether existing AI models could identify the protocol with the lowest measured response time.

2. Background

2.1 HTTP/1.1

HTTP/1.1 is an important version of the Hypertext Transfer Protocol and has been widely used for web communication. It supports persistent connections, which allow multiple requests and responses to use the same TCP connection.

However, HTTP/1.1 does not provide native multiplexing. When a website requires multiple resources, requests may need to wait for previous requests to complete. This can increase the time required to transfer web resources, especially when network latency is high.

2.2 HTTP/2

HTTP/2 was introduced to improve the efficiency of HTTP communication. One of its main features is multiplexing, which allows multiple streams to be transmitted through the same TCP connection. HTTP/2 also uses header compression to reduce the amount of repeated header information.

These features can reduce communication overhead and improve the performance of websites with multiple requests. However, HTTP/2 still uses TCP, so packet loss can affect the connection and may influence the performance of multiple streams.

2.3 HTTP/3 and QUIC

HTTP/3 is the latest major version of the HTTP protocol and uses QUIC as its transport protocol. Unlike HTTP/1.1 and HTTP/2, which use TCP, HTTP/3 uses QUIC over UDP.

QUIC provides connection management and reliable data transfer while supporting independent streams. This can reduce some of the limitations associated with TCP-based communication. HTTP/3 may therefore provide performance benefits under some network conditions.

However, the actual performance of HTTP/3 depends on the network environment, server configuration and workload. Therefore, experimental testing is still necessary to compare HTTP/3 with previous HTTP versions.

3. Methodology

3.1 Experimental Environment

The experiments were conducted using a web-based testing environment developed for HTTP performance analysis. The same website and server were used for all three HTTP protocols.

Microsoft Edge was used as the client browser for HTTP/3 testing. The browser was controlled through the Edge DevTools Protocol (CDP), which allowed the testing program to communicate with the browser and collect network information.

HTTP/3 testing was performed using QUIC support in Microsoft Edge. The same target website was used for HTTP/1.1, HTTP/2 and HTTP/3 to reduce differences caused by the website itself.

3.2 Website and Server Configuration

The experimental website is a personal web-based project containing several pages and static resources. The experiment focuses on a single HTML document request as the main workload.

The target document is the index.html page. The purpose of this design is to focus on the performance of a single HTTP document request and reduce the influence of additional webpage resources such as images, stylesheets and scripts.

The same target URL was used for HTTP/1.1, HTTP/2 and HTTP/3. This helps keep the requested content consistent during the comparison.

3.3 Experimental Procedure

The same testing procedure was used for HTTP/1.1, HTTP/2 and HTTP/3. Each protocol was tested 1000 times using the same target HTML document.

During each test, the testing program sent a request to the target website and recorded the network performance information. The collected data was saved as raw experimental data for later analysis.

A total of 3000 measurements were collected, including 1000 measurements for HTTP/1.1, 1000 measurements for HTTP/2 and 1000 measurements for HTTP/3.

After the experiments were completed, the three datasets were combined into one master dataset. The data was then organised and analysed using Microsoft Excel.

3.4 Performance Metrics

To evaluate the performance of different HTTP protocols, several important web performance metrics were selected.

3.4.1 Response Time

Response time represents the time required to complete the HTTP request and receive the response. It is used as one of the main metrics for comparing the performance of the three HTTP protocols. A lower response time indicates that the request was completed faster.

3.4.2 Time To First Byte

Time To First Byte (TTFB) represents the time between sending a request and receiving the first byte of the response. It can provide information about the initial response latency of the server and network connection. A lower TTFB generally indicates that the client receives the first response data more quickly. In this study, TTFB is used to compare the initial response performance of HTTP/1.1, HTTP/2 and HTTP/3.

3.4.3 Round Trip Time

Round Trip Time (RTT) represents the time required for a network packet to travel from the client to the destination and return. RTT is used to describe network latency during the experiment.

3.4.4 Jitter

Jitter represents the variation in network delay between different measurements. A higher jitter value indicates that network latency changes more between measurements, while a lower value indicates more stable delay. In this study, jitter is used to evaluate the stability of network delay during repeated measurements. Lower jitter indicates more consistent network conditions, while higher jitter indicates greater variation in delay.

3.4.5 Packet Loss

Packet loss represents the percentage of packets that do not successfully reach the destination. Packet loss can affect network communication and may influence the performance of HTTP protocols. A higher packet loss rate may result in retransmissions and additional communication delay. Therefore, packet loss is considered when interpreting differences in response time and network stability between the protocols.

3.4.6 Throughput

Throughput represents the amount of data transferred within a period of time. It is used to evaluate the data transfer performance of each HTTP protocol.

3.4.7 Data Size

Data size represents the amount of data transferred during the request. This metric helps describe the amount of information involved in each experiment.

3.4.8 Success Rate

Success rate represents the percentage of requests that completed successfully. It is calculated from the number of successful requests divided by the total number of requests.

3.5 AI-Assisted Analysis

A small AI-assisted analysis was conducted as a supplementary part of the study. Two AI models, ChatGPT and DeepSeek, were tested using the same experimental samples. The purpose was to examine whether the models could identify the HTTP protocol with the lowest measured response time.

In Round 1, the models were provided with RTT, jitter, packet loss, data size and throughput. In Round 2, additional RTT information, including average RTT, minimum RTT and maximum RTT, was provided. Response time was not provided to the models because it was used as the measured result for evaluating prediction accuracy.

For each round, 10 samples were provided to each model. The predicted protocol was compared with the actual protocol that achieved the lowest measured response time for the corresponding sample. Prediction accuracy was calculated as the number of correct predictions divided by the total number of samples.

4. Experimental Results

4.1 Response Time Analysis

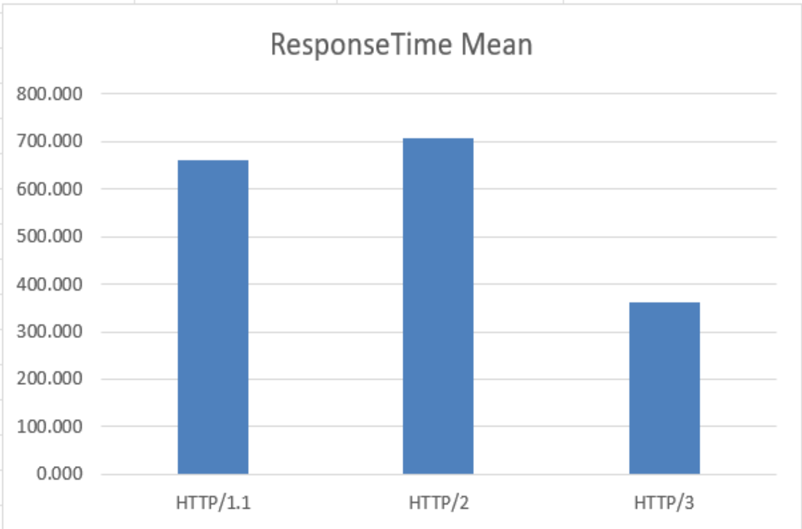

Figure 1. Response Time Comparison

Figure 1 shows the average response time of HTTP/1.1, HTTP/2 and HTTP/3 based on the experimental results.

HTTP/1.1 recorded an average response time of **661.281 ms**, while HTTP/2 recorded **707.736 ms**. HTTP/3 recorded **361.402 ms**. The average response time of HTTP/3 was substantially lower than those of HTTP/1.1 and HTTP/2 in the tested environment. This suggests that HTTP/3 provided a faster overall request completion time under the experimental conditions.

Among the three protocols, **HTTP/3** achieved the lowest average response time in this experiment. This result indicates that HTTP/3 achieved the fastest response under the tested conditions. The response time results should also be considered together with other network metrics, such as RTT, jitter and packet loss, because these factors can affect the observed performance.

However, the difference in response time may also be affected by network conditions and connection behaviour.

4.2 Time To First Byte (TTFB) Analysis

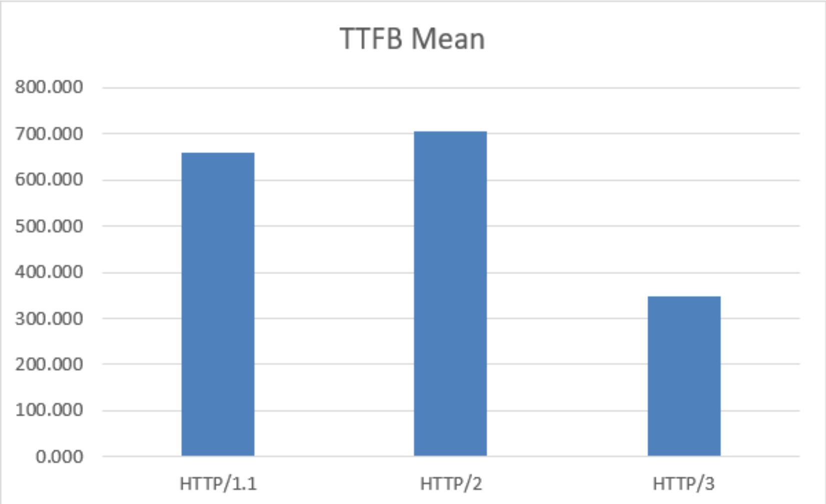

Figure 2. Average Time To First Byte (TTFB) Comparison

Figure 2 presents the average TTFB values for the three HTTP protocols.

HTTP/1.1 recorded an average TTFB of **660.703 ms**, followed by HTTP/2 with **706.627 ms** and HTTP/3 with **346.596 ms**.

The results show that **HTTP/3** achieved the lowest average TTFB. This suggests that the protocol provided a lower initial response delay under the tested conditions.

The TTFB results may not have the same ranking as response time because the two metrics describe different parts of the request process.

4.3 Network Performance Analysis

The network performance results were analysed using RTT, jitter and packet loss.

HTTP/1.1 recorded an average RTT of **192.307 ms**, while HTTP/2 recorded **193.572 ms** and HTTP/3 recorded **192.890 ms**.

The average jitter values were **3.225 ms**, **4.446 ms** and **4.244 ms** for HTTP/1.1, HTTP/2 and HTTP/3 respectively.

The packet loss results were **0%**, **0.760%** and **0.732%** respectively.

These results show that the network conditions were not exactly the same during every measurement. Therefore, network-related metrics are considered when discussing the differences between the three protocols.

The transferred data size was also recorded during the experiments. Although the same HTML document was requested, the recorded data size was not identical across all measurements and protocols. Therefore, the main differences in the results are discussed using response time, TTFB, network performance, throughput and success rate.

4.4 Success Rate

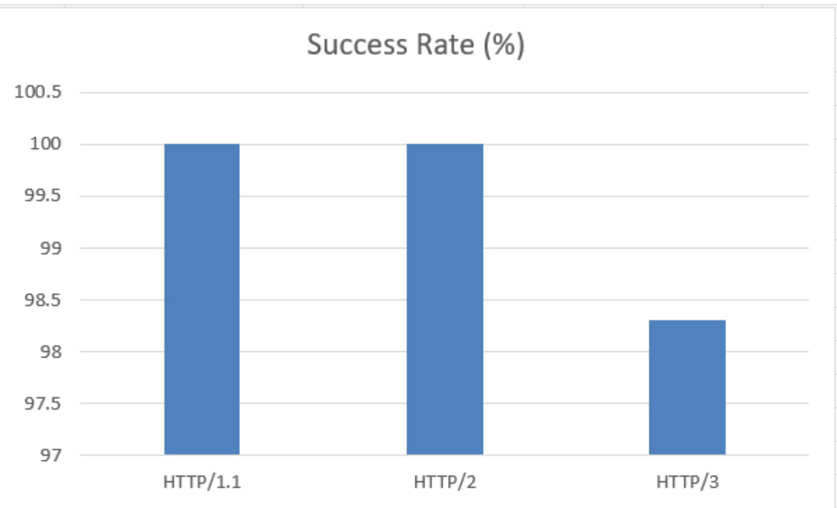

Figure 3. Success Rate Comparison

The success rate was calculated from the total number of requests and successful requests.

HTTP/1.1 achieved a success rate of **100%**, while HTTP/2 achieved **100%**. HTTP/3 achieved a success rate of **98.300%**.

The success rate provides additional information about the reliability of each protocol during the repeated experiments. A higher success rate indicates that more requests were completed successfully.

4.5 Statistical Summary

The main statistical results are summarised in Table 1.

Table 1. Statistical Summary of HTTP Performance

Metric	HTTP/1.1	HTTP/2	HTTP/3
Response Time Mean	661.281	707.736	361.402
Response Time Median	636	630	249
Response Time SD	489.372	955.447	258.295
TTFB Mean	660.703	706.627	346.596
TTFB Median	636	630	234
RTT Mean	192.307	193.572	192.890
Jitter Mean	3.225	4.446	4.244
Packet Loss Mean	0%	0.760%	0.732%
Throughput Mean	0.024	0.024	0.173
Success Rate	100%	100%	98.300%

Overall, HTTP/3 achieved the lowest response time and TTFB, while HTTP/1.1 and HTTP/2 achieved higher success rates in the experiment.

4.6 AI-Assisted Analysis

The AI-assisted experiment was conducted using 10 samples in each round. In Round 1, ChatGPT correctly identified the protocol with the lowest measured response time in 7 out of 10 samples, giving an accuracy of 70%, while DeepSeek achieved 40% accuracy.

In Round 2, ChatGPT correctly identified 6 out of 10 samples, giving an accuracy of 60%. DeepSeek also correctly identified 6 out of 10 samples, giving an accuracy of 60%.

Table 2. AI Prediction Accuracy Across Two Testing Rounds

Model	Round 1	Round 2
ChatGPT	70%	60%
DeepSeek	40%	60%

The results show that the AI predictions did not always match the measured experimental results.

5. Discussion

5.1 Comparison of HTTP Protocol Performance

The experimental results show that HTTP/1.1, HTTP/2 and HTTP/3 have different performance characteristics. HTTP/3 achieved the lowest average response time and TTFB in this experiment. HTTP/1.1 and HTTP/2 both achieved a 100% success rate, while HTTP/3 achieved 98.3%.

The results also show that the best protocol was not necessarily the same for every metric. Therefore, protocol performance should not be evaluated using only one metric. Response time, TTFB, RTT, jitter and packet loss provide different information about network performance.

5.2 Factors Affecting Experimental Results

Several factors may influence the experimental results.

First, network conditions may change during repeated experiments. Changes in latency, packet loss or network stability may affect response time and TTFB.

Second, server response time may affect the measured performance. Even when the same website is used, server conditions may change between different requests.

Third, browser and connection behaviour may also influence the results, particularly for HTTP/3 testing.

5.3 AI-Assisted Analysis

The AI-assisted results show that the two models produced different prediction performance. ChatGPT achieved 70% accuracy in Round 1 and 60% in Round 2, while DeepSeek improved from 40% to 60%. The results suggest that providing additional network metrics does not necessarily lead to higher prediction accuracy. The AI models can provide useful supplementary interpretations of network performance data, but their predictions should still be validated against direct experimental measurements.

6. Conclusion

This study presented an experimental comparison of HTTP/1.1, HTTP/2 and HTTP/3 using the same web environment and testing procedure. The experiment used 1000 measurements for each protocol, resulting in a total of 3000 measurements and providing a larger dataset for performance comparison.

The experiment measured response time, TTFB, RTT, jitter, packet loss, data size, throughput and success rate. The results provide a more detailed comparison of the performance characteristics of the three HTTP protocols.

The experimental results show that different HTTP protocols can perform differently depending on the selected metric and testing conditions. Therefore, evaluating HTTP performance using several metrics provides a more complete view than using a single measurement.

The AI-assisted analysis showed that AI models can provide useful supplementary interpretations of network performance data, but their predictions should still be validated against direct experimental measurements.

Catalog

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